Higher A

Marking guide • owner review

40 possible marks: 11 selected-response marks suitable for automatic checking; 29 written/working marks require structured self-marking or teacher review. These are original proposed schemes, not official AQA mark schemes.

Never award written scientific marks by keyword count. Accept equivalent valid science. Original submitted answers and review history must remain intact. A self-assessed total stays labelled self-assessed; a locally declared teacher mark is not authenticated external verification. No official GCSE grade follows from this topic paper.

1a · 1 mark · Selected response

Correct option: C. Ribosomes - protein synthesis

Ribosomes - protein synthesis.

Specification: 4.1.1.2 | Revision: O2

1b · 2 marks · Human/self-review

Chloroplast (1); site of photosynthesis / absorbs light for photosynthesis via chlorophyll (1).

Specification: 4.1.1.2 | Revision: O2

1c · 1 mark · Selected response

Correct option: A. A DNA loop not enclosed by a nucleus

A DNA loop not enclosed by a nucleus.

Specification: 4.1.1.1-4.1.1.2 | Revision: O2

1d · 1 mark · Selected response

Correct option: C. One

One order of magnitude (factor 10).

Specification: 4.1.1.1-4.1.1.2 | Revision: O2

2a · 1 mark · Selected response

Correct option: B. Centre first, then use fine focus

Centre the specimen first, then use fine focus at high power.

Specification: 4.1.1.2 RP01 | Revision: O4/O5

2b · 1 mark · Human/self-review

Distinguish structures/points that are closer together (1); making an image larger alone is insufficient.

Specification: 4.1.1.5 | Revision: O4

2c · 3 marks · Human/self-review

72 ÷ 24000 = 0.003 mm, or equivalent valid method (1); valid conversion from mm to m, ÷1000 (1); 3 × 10^-6 m in standard form (1). Accept 3.0 × 10^-6 m. Fully correct answer/unit can earn 3. A correct method/conversion can retain those marks despite arithmetic error. A correct non-standard-form decimal earns up to 2/3, because standard form is requested.

Specification: 4.1.1.5; MS 1b/3b | Revision: O4/O5

2d · 1 mark · Human/self-review

Use the known magnification/calibrated scale to calculate actual size, or include a correctly calibrated scale bar rather than treating displayed size as real size (1).

Specification: 4.1.1.2 RP01; WS 3.7 | Revision: O5

3a · 1 mark · Selected response

Correct option: A. Chromosomes normally occur in pairs and each carries many genes

Chromosomes normally occur in pairs and each carries many genes.

Specification: 4.1.2.1 | Revision: O6

3b · 2 marks · Human/self-review

DNA is replicated beforehand, producing copies of each chromosome (1); each new nucleus receives one copy of each chromosome / an identical complete chromosome set (1). Do not require cytoplasm division for this question about nuclei. Do not credit a claim that DNA merely grows bigger or that each new nucleus receives only half the original types of gene.

Specification: 4.1.2.2 | Revision: O6

3c · 1 mark · Selected response

Correct option: B. Many plant cells retain differentiation capacity throughout life; most animal types differentiate early

Many plant cells retain differentiation capacity throughout life; most animal types differentiate early.

Specification: 4.1.1.4 | Revision: O3/O6

3d · 2 marks · Human/self-review

Long extension carries signals over a long distance (1); branches connect with other cells/allow communication to several cells (1). Information about signal transmission is provided; do not require untaught synapse chemistry.

Specification: 4.1.1.3-4.1.1.4 | Revision: O3

4a · 2 marks · Human/self-review

Embryonic stem cells can form most human cell types (1); meristem stem cells can form any type of plant cell throughout plant life (1). Do not infer human-plant interchangeability.

Specification: 4.1.2.3 | Revision: O6

4b · 1 mark · Selected response

Correct option: B. The embryo has the same genes as the patient

The embryo has the same genes as the patient.

Specification: 4.1.2.3 | Revision: O6

4c · 2 marks · Human/self-review

Rapid production of many plants can increase numbers/preserve the existing useful resistance, supporting conservation in the short term (1); genetic uniformity can leave the clones susceptible to a different/new disease, so long-term survival is not guaranteed (1). Accept another defensible limitation based on the supplied evidence. A bare yes/no judgement earns no separate mark.

Specification: 4.1.2.3; WS 1.3 | Revision: O6

5a · 1 mark · Selected response

Correct option: C. More particles move down the gradient than up it overall

More particles move down the gradient than up it overall.

Specification: 4.1.3.1 | Revision: O7

5b · 2 marks · Human/self-review

P has surface area 1.5 mm² and volume 0.125 mm³, giving SA:V = 12:1 or an equivalent ratio (1); 12 ÷ 3 = four times Q (1). If P is wrong through arithmetic but a consistent valid comparison P÷3 is shown, allow only the comparison mark by error carried forward, for a maximum of 1/2. Do not credit a reversed comparison or invalid dimensions.

Specification: 4.1.3.1; MS 1c/5c | Revision: O8

5c · 2 marks · Human/self-review

Any two distinct links (1 each): thin leaf short diffusion path; air spaces create large internal area/short paths; stomata allow gas exchange with surrounding air. Do not require guard-cell regulation or later plant physiology.

Specification: 4.1.3.1 | Revision: O8

5d · 1 mark · Selected response

Correct option: C. The higher-temperature measurements consistently show a faster rate under these conditions

The higher-temperature measurements consistently show a faster rate under these conditions.

Specification: 4.1.3.1; WS 3.5 | Revision: O7

6a · 2 marks · Human/self-review

Continued net uptake is against the glucose concentration gradient and requires active transport (1); reduced respiration makes less energy available, so this uptake is likely to slow (1). Accept equivalent conditional wording. Do not infer that all respiration or all passive transport stops. No advanced cotransport mechanism is required.

Specification: 4.1.3.3 | Revision: O11

6b · 1 mark · Selected response

Correct option: C. Net movement of water from a dilute to a concentrated solution through a partially permeable membrane

Net movement of water from a dilute to a concentrated solution through a partially permeable membrane. Do not confuse solvent movement with solute movement.

Specification: 4.1.3.2 | Revision: O9

6c · 1 mark · Selected response

Correct option: C. Larger exchange area and steeper concentration gradient

Larger exchange area and steeper concentration gradient.

Specification: 4.1.3.1 | Revision: O7

7a · 2 marks · Human/self-review

(2.92 - 3.20) ÷ 3.20 × 100 = -8.75% (1 for valid method); -8.8% to one decimal place (1). A correct signed, rounded answer can earn 2. Accept 8.8% loss in words. -8.75% shows the correct method but misses the requested rounding, so 1/2. Do not double-penalise a sign error within the method.

Specification: 4.1.3.2 RP02; MS 1c | Revision: O9/O10

7b · 6 marks · Human/self-review

Human best-fit level marking, 0-6: estimate the zero near 0.32 mol/dm³ for this tissue/conditions; explain net water movement through partially permeable membranes; use the lack of replicates, unremoved surface liquid and varied time to question the universal/exact claim; propose justified controlled repeats, means and additional concentrations near zero. Do not demand knowledge of water-potential equations or an exact intracellular solute concentration. Explicitly credit comparing several plant tissue types across the concentration range under matched conditions (dimensions, time, temperature, solution volume and blotting), with repeats within each type and comparison of their estimated zero-change concentrations. Repeats of one tissue assess reliability but cannot establish the same zero-change concentration for every plant tissue; even agreement among several types cannot prove a universal claim.

Specification: 4.1.3.2 RP02; WS 2/3; MS 4a/4d | Revision: O9/O10

Six-mark best-fit rubric

0 marks: No relevant creditworthy content.

1-2 marks: A few relevant observations or improvements with little explanation; the judgement is weakly supported.

3-4 marks: Some linked use of the graph/osmosis and useful improvements, but the interpretation, justification or evaluation is incomplete.

5-6 marks: A coherent, evidence-based judgement links the zero-change estimate and net water movement to dataset limitations and a workable, controlled, repeated investigation that compares several tissue types under matched conditions and compares their zero-change estimates. Improvements are justified, and the limits of generalising to every plant tissue are explained.

Judge the whole response, select the best-fit level, then choose the lower/upper mark according to completeness and coherence. Do not count six keywords or require every possible control. For 5-6, both the scientific interpretation and justified methodological evaluation need to be developed. A strong answer can use other valid controls. For the top level, the response must develop how tissue-type comparisons under matched conditions address the universal claim; repeats of one tissue alone do not do so. Accept equivalent scientifically valid approaches and wording, including a justified counterexample design using another tissue type. Do not require a set phrase, named tissues, every listed control, or a checklist of points; apply the levels holistically. Merely saying “repeat for accuracy” is less developed than explaining that repeats expose variation and support a mean.

Illustrative model, not required wording

The joined graph crosses zero at about 0.32 mol/dm³. This is an estimate for this celeriac tissue and these conditions, not an exact value for every plant tissue. Net water movement through partially permeable membranes accounts for the gains and losses; movement in both directions continues at zero net change. One cylinder at each concentration gives no evidence of variation. Different times and surface solution confound the measured mass. I would use comparable equal-sized cylinders, fixed solution volume, time and temperature, and consistent blotting, then repeat each concentration and calculate means. Testing more concentrations close to zero would refine the estimate. Other tissues would need separate tests before any broader claim.

Foundation graph estimate: approximately 0.40 mol/dm³. Higher: approximately 0.32 mol/dm³; allow a justified graphical estimate from 0.31 to 0.33 on the supplied graph. These are estimates, not required spurious exactness. Percent changes in 7a are from separate pilot batches and are not answered by the 7b graph.